

STATISTICS 101 · ANSWER KEY - PROFESSOR COPY

Homework 1: What's Typical?

Grading guide. Total 21 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Mean and range

1. Five babies in the unit have heart rates of 140, 150, 130, 160, and 145 beats per minute. (a) What is the total? (b) What is the mean? (c) What is the range? (3 pts)

Answer: (a) 725. (b) $725 \div 5 = 145$. (c) $160 - 130 = 30$. 1 pt each.

2. Four babies weigh 900 g, 1,100 g, 1,000 g, and 1,000 g. What is the mean weight? What is the range? (2 pts)

Answer: Mean: $4,000 \div 4 = 1,000$ g. Range: $1,100 - 900 = 200$ g.

Part B — Score three babies with Apgar

3. Each baby's five Apgar signs have already been scored (0, 1, or 2 each). Add up the total, then write the category: 7-10 = doing well, 4-6 = may need some help, 0-3 = needs help right now. (6 pts)

Answer: Rosa: 9, doing well. Kai: 4, may need some help. Amara: 8, doing well. 1 pt per total, 1 pt per category.

Part C — Draw the data

4. On the chart above, draw a bar for each baby's heart rate from Part A: A = 140, B = 150, C = 130, D = 160, E = 145. Then answer: which baby's bar is tallest, and which bar is closest to the mean? (3 pts)

Answer: Five bars at the right heights (2 pts; 1 pt off for two or more wrong heights). Tallest: D (160). Closest to the mean of 145: E (145) — 1 pt.

5. Two units both report an average Apgar of 7. What does the average alone NOT tell you? (1 pt)

Answer: A. Whether the scores were all close to 7 or a mix of very high and very low. - The mean hides the spread — that's why we also look at the range.

Part D — Flashback

6. From NICU Tutor: what is the normal heart rate range for a newborn, in beats per minute? (1 pt)

Answer: About 100-160 (sometimes higher when upset).

7. From NICU Tutor: what does the word bradycardia mean? (1 pt)

Answer: A heart rate that is too slow ('brady' = slow, 'cardia' = heart).

8. From Calculus: rate of change = $\frac{\text{change}}{\text{time}}$. (1 pt)

Answer: Change $\div$ time.

Part E — Think like a doctor

9. A new nurse says, “This baby's Apgar was 5 at one minute, so he'll probably have problems growing up.” What would Dr. Virginia Apgar herself have said? (Think about what the score is FOR.) (3 pts)

Answer: The Apgar score tells the team how the baby is doing RIGHT NOW and whether to help — it is not a prediction of the child's future (1 pt). Many babies score low at one minute and fine at five minutes once they start breathing well (1 pt). Apgar warned against using her score to predict a child's future or judge the mother or the birth (1 pt).